

Case Study: Building a Peak-Hour Decision Support System for Dark Store Managers

Background & Context

Quick commerce in India has grown very fast in the last few years. In metro cities, customers now expect groceries and essentials within 10–15 minutes.

Companies like Blinkit, Zepto, and Swiggy Instamart operate hundreds of dark stores across India.

For example:

- Blinkit operates 500+ dark stores
- Zepto operates 300+ dark stores
- Swiggy Instamart operates 600+ dark stores
- Together, the industry runs over 1,500–2,000 dark stores across major cities

Industry estimates suggest:

- India's quick commerce market is expected to cross \$5 billion in the next few years
- Evening peak hours (6–10 PM) can see 30–50% higher order volume than normal hours
- A high-demand dark store may process 1,200–2,000 orders per day

- SKU count per store can range between 3,000–5,000 items

As customer expectations rise, delivery time promises have become tighter. Even a delay of a few minutes can affect ratings and repeat usage.

Across many cities, during peak hours, the following patterns are observed:

- Delivery delays increase
- Wrong or missing items increase
- Refunds and compensations rise
- Rider waiting time near stores increases
- Store staff work extended shifts

For example, during rain or IPL match evenings:

- Order volume can spike sharply within 20–30 minutes
- Demand for snacks, beverages, and essentials increases suddenly
- Picking errors rise due to rush
- Rider clustering happens outside stores

These issues appear as customer-facing problems.

But most of them start inside the dark store.

Dark stores handle:

- Order picking
- Inventory tracking

- Staff allocation
- Rider handoff
- Packing speed

Most stores operate within fixed space and limited manpower.

When order volume increases sharply inside the same physical setup:

- Picking takes longer
- Riders wait longer
- Orders miss promised timelines
- Refund cost increases
- Customer complaints rise

Even a 5% drop in on-time delivery in high-demand zones can impact hundreds of orders daily across multiple stores.

While total revenue may continue to grow, repeated peak-hour instability can:

- Increase cost per order
- Increase refund payouts
- Increase manager stress and attrition
- Reduce long-term customer trust

Dark stores are not just fulfillment points.

They are the operational core of quick commerce.

If they struggle during peak hours, the entire system feels the impact.

Problem Statement

During peak hours, Blinkit's high-demand dark stores experience operational chaos due to decision overload and limited real-time clarity for managers.

Managers:

- React instead of anticipate
- Struggle to prioritize issues
- Lack a single clear control layer
- Have limited visibility into peak stress signals

The system functions during average load.

It destabilizes under peak pressure.

Blinkit needs a product that helps dark store managers make fast, informed decisions during peak hours.

The Challenge

Design a product that enables Blinkit's dark store managers to:

- Detect operational stress early
- Understand what needs attention first
- Take quick corrective actions
- Maintain delivery performance under peak conditions

You are not redesigning the supply chain.

You are not solving demand forecasting.

You are solving manager decision clarity during peak hours.

Your Role as a Product Manager

You are leading the design of a Peak-Hour Decision Support Product for dark store managers at Blinkit.

Your responsibilities include:

- Clearly defining the manager's core decision problem
 - Identifying what information is missing or unclear
 - Designing a focused product solution
 - Ensuring usability under stress
 - Defining measurable success criteria
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Deliverables (What You Must Submit)

Your submission must include:

1. A clear problem definition
2. Peak-hour workflow mapping
3. Identification of the core decision gap
4. Defined primary user persona (manager profile)

5. Key metrics to improve (e.g., SLA, picking time, stress signals)
 6. Proposed product concept
 7. One bold “moonshot” idea
 8. 2–3 practical supporting features
 9. High-level product flow
 10. MVP scope definition
 11. Trade-offs and limitations
 12. Basic wireframes or working prototype
 13. Keep it realistic and focused.
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Approach to Tackling the Problem

1 Discovery Phase (40 Points)

You must:

- Observe and map peak-hour manager workflow
- Identify when instability begins
- Define the core decision bottleneck
- Separate symptoms from root cause
- Understand what managers can and cannot control

Avoid assuming the solution too early.

2 Solution Phase (40 Points)

You must:

- Propose a clear product concept
- Define how it reduces peak-hour chaos
- Align directly with identified problem
- Include one bold idea
- Include practical, implementable features
- Explain feasibility and trade-offs

Focus on impact, not feature count.

3 Design Phase (20 Points)

You must:

- Design for simplicity
- Show before-and-after workflow
- Ensure low cognitive load
- Prioritize clarity over complexity

Peak-hour design must be glanceable and action-oriented.

Constraints

- Cannot reduce growth demand
- Cannot drastically reduce SKU count
- Cannot expand physical store size
- Must work within realistic operational limits
- Must be scalable across multiple stores

Scoring Criteria

Category	Criteria	Points
Discovery	Clear problem definition	15
Discovery	Understanding of manager reality	10
Discovery	Identification of core decision gap	10
Discovery	Logical clarity	5
Solution	Strong alignment with problem	15
Solution	Practical and feasible approach	10

Solution	One bold idea	10
Solution	Clear trade-offs	5
Design	Simplicity under stress	10
Design	Clear workflow	5
Design	Focused design	5

Important Note

The above info and scoring table is only directional.

You are free to structure your PRD using your own headers and framework, as long as the logic is clear and contextual.